AN ASSIGNMENT

ON

# COMPUTER HARDWARE

**(CSC 104)**

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## QUESTION

1. Write about micro processor and RAM

2. Write on different types of processors and their functions

## ANSWER

## QUESTION 1

**Microprocessor**

A microprocessor is the central processing unit (CPU) of a computer, responsible for executing instructions and performing calculations. It is a small but powerful chip that acts as the “brain” of a computer, controlling all operations.

The processor, also known as the CPU, provides the instructions and processing power the computer needs to do its work. A microprocessor is a computer’s central processing unit (CPU) that is contained on a single integrated circuit (IC). It is also known as a micro processing unit (MPU).

Microprocessors are the “brain” of computers and many electronic devices. They accept binary data as input, process it, and provide results in binary form. They contain arithmetic, logic, and control circuitry. They interpret and execute program instructions. They perform arithmetic operations




**FIG 1: An example of a microprocessor**

**Functions of microprocessor**

1. Receiving data from the user or from another device
2. Performing arithmetic and logic operations on data
3. Temporarily storing data in the microprocessor's memory
4. Sending data to the user or to another device
5. Executing programs stored in memory
6. Controlling the activities of the computer system
7. Retrieves commands from memory.
8. Interprets what needs to be done.
9. Performs calculations and logical operations.
10. Saves data temporarily or permanently.

**Types of microprocessor**

1. CISC (Complex Instruction Set Computer) – Uses a large set of complex instructions to execute multiple tasks per instruction.
2. RISC (Reduced Instruction Set Computer) – Uses a simplified instruction set for faster execution and efficiency.
3. Superscalar Microprocessor – Executes multiple instructions simultaneously to improve performance.
4. DSP (Digital Signal Processor) – Specialized for processing audio, video, and image data in real time.
5. Embedded Microprocessor – Designed for specific applications like smart devices, ATMs, and industrial systems.

6. GPU (Graphics Processing Unit) – Optimized for parallel processing in graphics rendering and AI computations.
7. AI and NPU (Neural Processing Unit) – Built for artificial intelligence, deep learning, and machine learning tasks.
8. Multi-core Microprocessor – Contains multiple processing units (cores) to enhance multitasking and speed.

**RAM**

RAM is a common computing acronym that stands for random-access memory. Sometimes it's called PC memory or just memory. In essence, RAM is your computer or laptop's short-term memory. It's where the data is stored that your computer processor needs to run your applications and open your files.

RAM stands for Random Access Memory, and is used as a short-term memory storage space. It is a type of computer memory that stores data and machine code currently being used. It allows fast access to data, enabling smooth multitasking and quick program execution. Unlike a hard drive or SSD, RAM is volatile memory, meaning it loses all data when the computer is turned off.

It is a computer's short-term memory that stores data for quick access by the CPU. RAM is essential for a computer's performance, especially for multitasking. It is volatile, meaning it loses its contents when the power is turned off. It is  faster to read from and write to than secondary storage, such as hard disk drives (HDDs) or solid-state drives (SSDs).



**FIG 2: Example of a RAM**

**Functions of RAM**

1. Keeps data readily available for the CPU.
2. Reduces lag and improves responsiveness.
3. Allows multiple applications to run simultaneously.
4. RAM stores data for the operating system, software programs, and other data in use.
5. RAM allows users to run multiple applications at the same time.
6. RAM allows users to open larger files and use more demanding applications.
7. RAM allows apps to run more smoothly by providing quick access to data.

**Types of RAM**

1. SRAM (Static RAM) – Faster and more reliable but expensive; used for cache memory.
2. DRAM (Dynamic RAM) – Slower than SRAM and needs constant refreshing; used as main memory in computers.
3. SDRAM (Synchronous DRAM) – Synchronized with the CPU clock for improved performance.
4. DDR (Double Data Rate) RAM – Transfers data twice per clock cycle, with versions like DDR1, DDR2, DDR3, DDR4, and DDR5, each improving speed and efficiency.
5. GDDR (Graphics DDR) RAM – Specialized for GPUs to handle high-speed graphics processing; versions include GDDR3, GDDR5, GDDR6, etc.
6. VRAM (Video RAM) – Used in graphics cards to store image and video data for display.
7. LPDDR (Low Power DDR) RAM – Energy-efficient version of DDR used in mobile devices and laptops.
8. ECC (Error-Correcting Code) RAM – Detects and corrects memory errors, used in servers and critical computing applications.

**Similarities Between Microprocessor and RAM**

1. The CPU processes data, while RAM ensures quick access to it.
2. RAM stores temporary data for active programs, while the CPU temporarily processes and manipulates data before storing or discarding it.
3. A powerful CPU and sufficient RAM together determine the system's speed and responsiveness.
4. RAM feeds the CPU with data, and the CPU processes it before returning results.
5. They are physical parts of a computer, installed on the motherboard.

**Differences Between Microprocessor and RAM**

1. The microprocessor executes instructions and processes data, while RAM temporarily stores data for quick access by the CPU.
2. The microprocessor is the processing unit (the brain of the computer), whereas RAM is a memory unit used for temporary storage.
3. The microprocessor does not store data permanently; it only processes it. RAM, on the other hand, loses all stored data when the computer is turned off because it is volatile memory.
4. The microprocessor is faster because it performs complex calculations and logical operations. RAM is slower than the CPU but much faster than permanent storage like a hard drive or SSD.
5. The microprocessor can function independently but needs RAM for quick access to data. RAM, however, is completely dependent on the CPU, as it only stores temporary data for processing.

6. The microprocessor is installed in a dedicated socket on the motherboard, while RAM is inserted as removable memory modules (RAM sticks) on the motherboard.
7. The microprocessor has limited upgrade options, often requiring a full replacement to improve performance. RAM is easier to upgrade by simply adding more memory modules.

**How Microprocessors and RAM Work Together**

When you run a program, the microprocessor fetches instructions from storage (hard drive or SSD) and loads them into RAM. The CPU then processes these instructions using the data stored in RAM. More RAM means the processor can work faster without waiting for data to be retrieved from slower storage.

## QUESTION 2

**Different Types of Processors and Their Functions**

1. CISC (Complex Instruction Set Computer) Processors: Used in desktop computers and servers. Examples are Intel x86 series, AMD Ryzen.

Function: Executes complex instructions in fewer steps, reducing the need for software optimization.

2. RISC (Reduced Instruction Set Computer) Processors: Used in smartphones, embedded systems, and gaming consoles. Examples are ARM processors (Apple M-series, Qualcomm Snapdragon).

Function: Uses a simplified set of instructions to execute tasks faster and more efficiently.

3. Superscalar Processors: Used in High-performance computing, gaming PCs, and workstations. Example are Intel Core i9, AMD Ryzen 9.

Function: Can execute multiple instructions simultaneously in a single cycle, improving performance.

4. DSP (Digital Signal Processing) Processors: Used in Mobile phones, digital cameras, and medical devices. Examples are Texas Instruments TMS320, Qualcomm Hexagon DSP.

Function: Optimized for real-time audio, video, and image processing.

5. Embedded Processors: Used in smart appliances, IoT devices, and industrial automation. Examples are ARM Cortex-M series, Intel Quark.

Function: Designed for specific tasks in embedded systems, providing power efficiency and reliability.

6. GPU (Graphics Processing Unit) Processors: Used in Gaming, 3D modeling, machine learning, and video editing. Examples are NVIDIA GeForce, AMD Radeon.

Function: Specializes in parallel processing for rendering images, videos, and AI computations.

7. AI and Neural Processing Unit (NPU) Processors: Used in AI-powered devices, self-driving cars, and smartphones. Examples are Google Tensor, Apple Neural Engine, Huawei Ascend.

Function: Handles artificial intelligence tasks, including deep learning and machine learning acceleration.

8. Multi-core Processors: Used in Laptops, desktops, and high-performance computing. Examples are Intel Core i7 (Quad-core), AMD Ryzen 9 (Octa-core).

Function: Contains multiple cores to handle several tasks simultaneously, improving multitasking and performance.